13.1.1 Exercises - Newton's law of gravitation

1. According to the Newton's law of gravitation, where G is the gravitational constant.
   a. Rearrange the equation to make G its subject.

   b. what the units of G in SI base units.

2. There are two objects, A and B. The mass of A is greater than the mass of B. A and B attract each other with a gravitational force. These forces are F_1 and F_2:
   a. What can you say about the directions of forces F_1 and F_2?

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   b. What can you say about the magnitudes of forces F_1 and F_2?

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   c. A and B are initially stationary. The gravitational forces cause them to accelerate towards each other. Which object will have the greater acceleration? Explain your answer.

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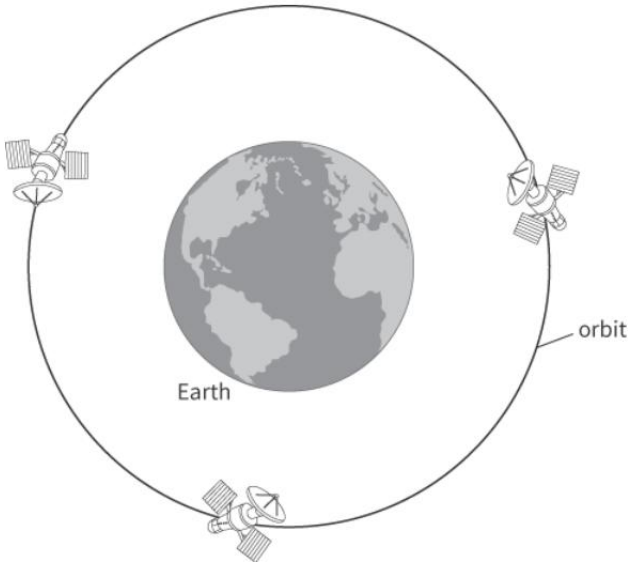
3. Calculate the gravitational force of the Earth on a mass of 6.0kg placed on its surface (mass of Earth = 6.0×10^{24} kg; radius of Earth = 6400 km).

4. Calculate the gravitational force of the Sun on the Earth (mass of Sun = 2.0×10^{30} kg; radius of Earth's orbit around Sun = 150×10^{6} km).

5. This diagram shows three similar spacecraft that share a circular orbit around the Earth. Each spacecraft has a mass of 450 kg:



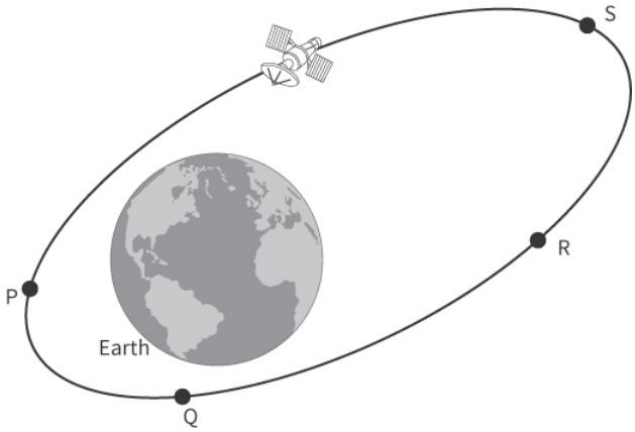
a. Add a force arrow to each spacecraft to show the force of the Earth's gravitational pull on it.

b. Each spacecraft orbits at a distance of 2600 km above the Earth's surface. Calculate the distance of each spacecraft from the centre of the Earth and the gravitational force on it.

c. A spacecraft must travel at the correct speed if it is to stay in its orbit. Calculate this speed for the spacecraft.

d. Calculate the time for one of the spacecraft to complete a single orbit of the Earth.Give your answer in minutes.

6. This diagram shows a spacecraft in an elliptical orbit around the Earth. Its distance from the Earth varies as it travels around its orbit:



Include reasons with your answers to these questions:

a. State the point at which the Earth's gravitational pull on the spacecraft is weakest.

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b. State the point at which the spacecraft's gravitational potential energy is greatest.

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c. The spacecraft's total energy (KE+PE) is constant. State the point at which the spacecraft will be moving most slowly.

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7. A spacecraft is in a geostationary orbit and orbits a planet once each day. It appears to remain at a fixed point in the sky, above the planet's equator.

a. Determine the period of orbit of a geostationary spacecraft around the Earth. Give your answer in seconds.

b. Calculate the radius of orbit for a geostationary planet orbiting Mars. Mars takes 24.6 hours to complete one rotation around the Sun. (mass of Mars: 6.42×10^{23} kg)

8.

a. State Newton's law of gravitation.

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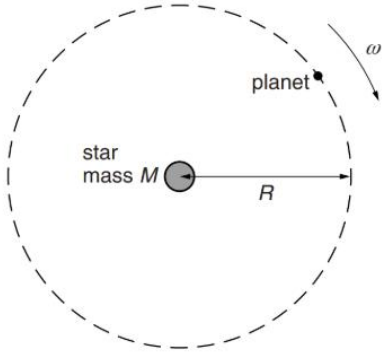
b. A star and a planet are isolated in space. The planet orbits the star in a circular orbit of radius R



The angular speed of the planet about the star is ω.

By considering the circular motion of the planet about the star of mass M, show that ω and R are related by the expression

$$R^3\omega^2 = GM$$

where G is the gravitational constant. Explain your working. [3]

c. The Earth orbits the Sun in a circular orbit of radius 1.5×10^8 km. The mass of the Sun is 2.0×10^{30} kg.

A distant star is found to have a planet that has a circular orbit about the star. The radius of the orbit is 6.0×10^8 km and the period of the orbit is 2.0 years.

Use the expression in (b) to calculate the mass of the star.

mass = ... kg [3]